

No. of Functional Features Included in the Solution

Date	15 March 2026
Team ID	
Project Name	Smart Lender:AI/ML-Based Loan Approval Prediction System

Functional Features Overview:

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Applicant Data Entry	Allows users to enter loan applicant details through a web form.	User Interface (Flask + HTML)	Done	High
2	Data Preprocessing	Cleans and prepares input data before prediction.	Data Processing	Done	High
3	Loan Approval Prediction	Predicts loan approval using the trained XGBoost model.	Machine Learning Model	Done	High
4	Multiple Model Evaluation	Trains and compares Decision Tree, Random Forest, KNN, and XGBoost models.	Model Training	Done	Medium
5	Model Loading	Loads the saved machine learning model for real-time predictions.	Model Deployment	Done	High
6	Prediction Result Display	Displays loan approval or rejection instantly on the web page.	Flask Application	Done	High
7	Responsive Web Interface	Provides an easy-to-use interface accessible	Frontend (HTML/CSS/Bootstrap)	Done	Medium

		through a web browser.			
8	IBM Cloud Deployment Support	Supports deployment of the Flask application on IBM Cloud.	Cloud Deployment	In Progress	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Applicant Data Entry Form, Prediction Result Page
2	Backend / Logic	4	Data Preprocessing, Model Loading, Loan Prediction, Flask Routing
3	Database / Storage	1	Pickle Model Storage (model.pkl)
4	API / Integration	1	Flask Integration with Machine Learning Model
5	Security / Authentication	0	Not Implemented